

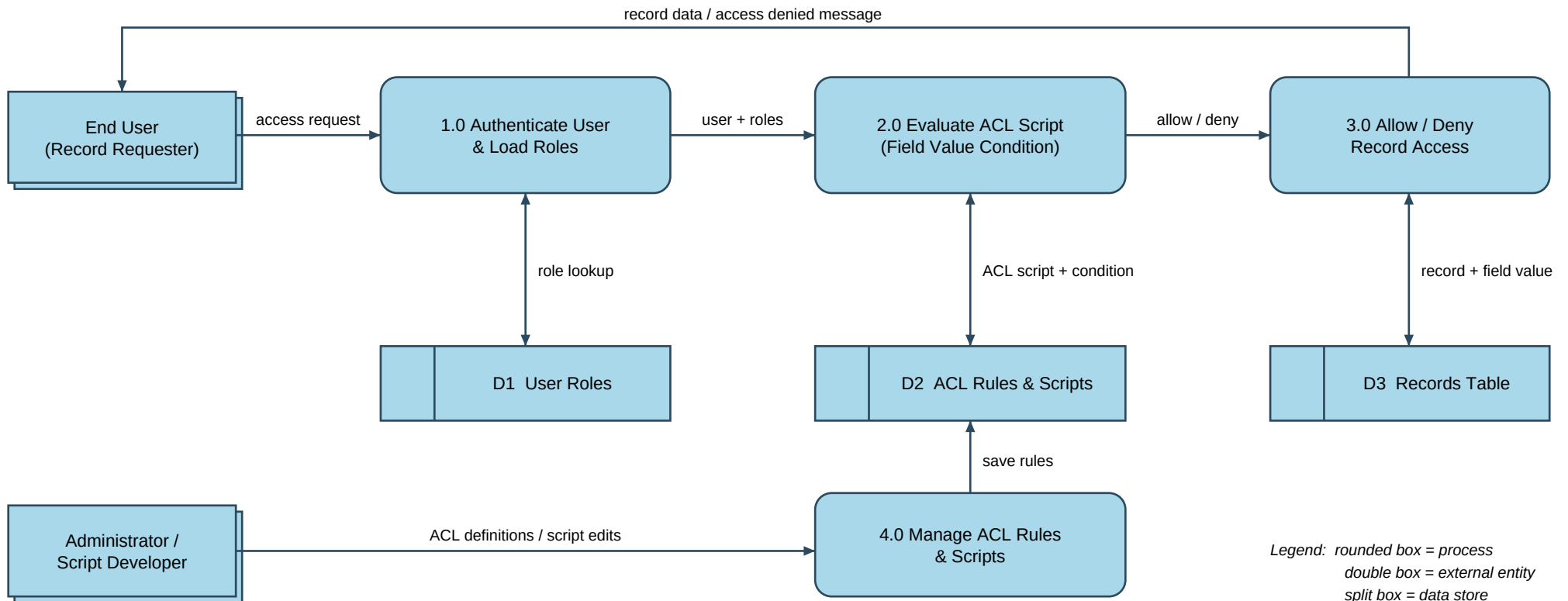
Project Design Phase-II

Data Flow Diagram & User Stories

Date	02 October 2026
Team ID	SWTID-2026-1268
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	4 Marks

Data Flow Diagram (Level 0)

A Data Flow Diagram shows how data enters and leaves the system, what changes the information, and where data is stored. The DFD below shows how a record request is checked by a script-controlled Access Control List (ACL) that compares a field value on the record with the requesting user's attributes before access is granted.



Flow Description

1. The end user signs in and requests a record (list, form or API call).
2. The system authenticates the user and loads their roles and attributes from **D1 User Roles**.
3. The ACL engine fetches the matching ACL rule and its script/condition from **D2 ACL Rules & Scripts**.
4. The script reads the field value of the target record (for example, department, region, or state) from **D3 Records Table** and compares it with the user's roles/attributes, returning **true** or **false**.
5. If the script returns true, the record data is displayed; if false, the record is hidden or an "access denied" message is returned.
6. The administrator / script developer creates, tests and updates ACL rules and scripts through process 4.0, which stores them in D2.

User Stories

The table below lists the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Administrator	Authentication & Roles	USN-1	As an administrator, I can log in securely with my credentials.	I can access the ACL configuration console.	High	Sprint-1
		USN-2	As an administrator, I can create roles and assign them to users.	Assigned roles appear on the user's profile.	High	Sprint-1
	ACL Configuration	USN-3	As an administrator, I can create an ACL on a table and choose the field that controls access.	The ACL is saved and listed against the table.	High	Sprint-1
		USN-4	As an administrator, I can define the field-value condition (e.g. department = HR) that decides who may open a record.	The condition is saved and validated.	High	Sprint-1
		USN-5	As an administrator, I can enable or disable an ACL without deleting it.	A disabled ACL is no longer enforced.	Medium	Sprint-2
Script Developer	Script-Controlled ACL	USN-6	As a script developer, I can write an ACL script that returns true/false using the current record's field value and the current user.	The script runs without errors and returns a boolean.	High	Sprint-2
		USN-7	As a script developer, I can test the script against sample records before activating it.	Test shows allow/deny result for each sample record.	Medium	Sprint-2
		USN-8	As a script developer, I can reuse common logic through a shared script include.	Multiple ACLs call the same function successfully.	Low	Sprint-3
Standard User (Web)	Authentication	USN-9	As a user, I can log in with my email and password.	I can access my dashboard.	High	Sprint-1
	Record Access	USN-10	As a user, I can view the list of records and see only those I am permitted to access.	Records that fail the ACL script are not listed.	High	Sprint-2
		USN-11	As a user, I see a clear "access denied" message if I open a restricted record directly through its URL.	No restricted data is shown; message is displayed.	High	Sprint-2
		USN-12	As a user, I can view sensitive fields on a record only when my role allows it.	Restricted fields are hidden or masked for other users.	Medium	Sprint-3

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Standard User (Mobile)	Record Access	USN-13	As a mobile user, I can view my permitted records with the same access rules as the web.	Mobile results match web results for the same user.	Low	Sprint-4
Security Auditor	Audit & Reporting	USN-14	As an auditor, I can view a log of allowed and denied record access attempts.	Log shows user, record, time and result.	Medium	Sprint-3
		USN-15	As an auditor, I can export the access log as a report.	A downloadable report is generated.	Low	Sprint-4